Process & Decision Documentation

Project/Assignment Decisions

Add music to the sketch and player trail path

GenAI Documentation

Human Decision Point(s)
- I chose the music to add to my sketch
- I decided to the player trail path as visual effects
- I got rid of the planets gravity effects (it was making the sketch too complicated visually) and the star sound.

Transcript

Add sound or simple physics to your sketch e.g., visual ehects triggered by motion, collisions, or player actions.

Here's a **simple way to add both physics and sound** to your galaxy experience while keeping the meditative feel.

We'll add:

> ⭐ **Sound when discovering stars**
>
> 🪐 **Planet gravity (soft pull when near planets)**
>
> ✨ **Particle trail when moving**

This creates a **reactive multi-sensory experience**.

1️⃣ Add Sound (in `sketch.js`)

At the top:

```
let starSound;
```

In `preload()`:

```
function preload() {
  worldData = loadJSON("world.json");
  starSound = loadSound("star.wav"); // soft chime sound
}
```

In `setup()` add:

```
userStartAudio(); // enables sound after interaction
```

2️⃣ Add Particle Motion Effect

Modify the **Player class**.

Add this in the constructor:

```
this.trail = [];
```

Add this at the end of `updateInput()`:

```
// add trail particle
if (abs(this.vx) + abs(this.vy) > 0.3) {
  this.trail.push({
    x: this.x,
    y: this.y,
    life: 40
  });
}

if (this.trail.length > 80) this.trail.shift();
```

Modify `draw()`:

```
draw() {

  // trail particles
  noStroke();
  for (let p of this.trail) {
    fill(180, 220, 255, p.life * 2);
    ellipse(p.x, p.y, 6);
    p.life--;
  }

  // glow
  for (let i = 40; i > 0; i -= 8) {
    fill(120, 180, 255, 20);
    ellipse(this.x, this.y, i);
  }

  fill(200, 230, 255);
  ellipse(this.x, this.y, 14);
}
```

Now movement leaves a **soft cosmic trail**.

3️⃣ Add Planet Gravity (Simple Physics)

Inside `WorldLevel.drawWorld(player)` add this **before drawing planets**:

```
for (const p of this.planets) {

  let d = dist(player.x, player.y, p.x, p.y);

  if (d < p.size * 2) {
    let force = map(d, p.size * 2, p.size, 0.02, 0.08);

    let angle = atan2(p.y - player.y, p.x - player.x);

    player.vx += cos(angle) * force;
    player.vy += sin(angle) * force;
  }
```

```
}
```

This creates **gentle gravitational pull near planets**.

4️⃣ Play Sound When Discovering Stars

In the **hidden star section** replace:

```
if (d < 60 && !hs.found) {
  hs.found = true;
}
```

with:

```
if (d < 60 && !hs.found) {
  hs.found = true;
  if (starSound) starSound.play();
}
```

Now **discovering a star triggers a soft chime**.

🌌 Final Result

Your galaxy now has:

⭐ Hidden stars → **sound feedback**
🪐 Planets → **gravity physics**
✨ Movement → **particle trails**
🎥 Camera → **smooth cinematic drift**

The experience becomes **reactive and multi-sensory** without losing the calm atmosphere.

If you want, I can also show you **one small addition that makes this look 10× more professional**:

nebula clouds that slowly move across the galaxy (very easy but visually powerful).

there' no sound when the stars are discovered. The player disappeared mid game, why?

Both issues are common with **p5.js + p5.sound** and the **gravity code** we added. Here's what's happening and how to fix it.

1️⃣ Why there is no sound

Modern browsers **block audio until the user clicks the page**.
Even if you load the sound correctly, it won't play until audio is started by interaction.

Fix

Add this to your sketch:

```
function mousePressed() {
  userStartAudio();
}
```

Then keep your discovery code:

```
if (d < 60 && !hs.found) {
  hs.found = true;

  if (starSound && !starSound.isPlaying()) {
    starSound.play();
  }
}
```

Also make sure:

You imported the sound library in **index.html**:

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/
1.9.0/p5.min.js"></script>
```

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/
1.9.0/addons/p5.sound.min.js"></script>
```

Without `p5.sound`, audio will never work.

2️⃣ Why the player disappeared

The **planet gravity** can accelerate the player indefinitely.
When velocity grows too large, the player shoots off screen instantly.

Fix: limit velocity

Inside the **Player.updateInput()** add this at the end:

```
// limit max speed
this.vx = constrain(this.vx, -4, 4);
this.vy = constrain(this.vy, -4, 4);
```

3️⃣ Add friction so gravity doesn't break the game

Also ensure this exists (important):

```
this.vx *= 0.92;
this.vy *= 0.92;
```

This keeps motion **stable and floaty**.

4️⃣ Optional safety clamp (recommended)

Add this in `draw()` after updating the player:

```
player.x = constrain(player.x, 0, level.w);
player.y = constrain(player.y, 0, level.h);
```

This prevents leaving the world entirely.

🌌 After these fixes

Your game will have:

- ⭐ **Star discovery sound**
- 🪐 **Stable gravity physics**
- 🎮 **Player that never disappears**

💡 If you'd like, I can also give you a **very small 10-line improvement** that makes the galaxy feel **much more alive**:
planets that slowly orbit tiny moons.

where does the sound code go

Here is **exactly where each sound piece goes** in your files.

1️⃣ Top of sketch.js

Add the sound variable **with your other globals**.

```
const VIEW_W = 800;
const VIEW_H = 480;

let worldData;
let level;
let player;

let camX = 0;
```

```
let camY = 0;

let starSound;   // ⭐ ADD THIS
```

2️⃣ Inside preload()

Load the sound file.

```
function preload() {
  worldData = loadJSON("world.json");
  starSound = loadSound("star.wav");   // ⭐ ADD THIS
}
```

Make sure `star.wav` is in your project folder.

3️⃣ Inside setup()

Add this line **at the bottom of setup()**.

```
function setup() {
  createCanvas(VIEW_W, VIEW_H);
  textFont("sans-serif");
  textSize(14);

  userStartAudio(); // ⭐ ADD THIS

  level = new WorldLevel(worldData);

  const start = worldData.playerStart ?? { x: 300, y: 300,
speed: 3 };
  player = new Player(start.x, start.y, start.speed);

  camX = player.x - width / 2;
  camY = player.y - height / 2;
```

```
}
```

4️⃣ Add this function anywhere in sketch.js

Usually placed **below draw()**.

```
function mousePressed() {
  userStartAudio();
}
```

This lets the browser enable sound when the player clicks.

5️⃣ Inside WorldLevel.drawWorld()

Find this part of your hidden star code:

```
if (d < 60 && !hs.found) {
  hs.found = true;
}
```

Replace it with:

```
if (d < 60 && !hs.found) {
  hs.found = true;

  if (starSound && !starSound.isPlaying()) {
    starSound.play();
  }
}
```

6️⃣ IMPORTANT (HTML file)

Make sure your **index.html** includes the sound library:

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/1.9.0/p5.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/1.9.0/addons/p5.sound.min.js"></script>
```

Without this, sound will never work.

🌌 Final Flow

1️⃣ Player moves near hidden star
2️⃣ Star becomes discovered
3️⃣ `starSound.play()` runs
4️⃣ Soft chime plays

If you'd like, I can also show you a **very cool 5-line trick** to make the **sound pitch change depending on how fast the player moves**, which makes the galaxy feel **much more alive**.

forget about the chip sound for the stars, I just wanna add a sound in the game

The easiest way is to add **ambient space music** that plays in the background. This keeps the meditative feeling of your galaxy.

Here's the **simplest working setup**.

1️⃣ Add the sound library (index.html)

Make sure this line is included:

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/1.9.0/addons/p5.sound.min.js"></script>
```

2️⃣ Add a sound variable (top of `sketch.js`)

Put it with your other variables.

```
let spaceMusic;
```

3️⃣ Load the sound (inside `preload()`)

```
function preload() {
  worldData = loadJSON("world.json");
  spaceMusic = loadSound("space.mp3");
}
```

Put **space.mp3** in your project folder.

4️⃣ Start the sound when the player clicks

Add this **anywhere in sketch.js**:

```
function mousePressed() {

  userStartAudio();

  if (!spaceMusic.isPlaying()) {
    spaceMusic.setVolume(0.5);
    spaceMusic.loop();
  }

}
```

🎮 What happens now

Player opens the game

Player clicks anywhere

Ambient **space music starts looping**

The galaxy exploration now has sound.

✅ This satisfies the assignment because your game now includes **sound reacting to player input (mouse click)**.

If you want something **cooler but still simple**, I can also show you a **10-line upgrade** where:

music **gets louder when you move faster**

music **gets quieter when you stop**

It makes the experience feel **alive and interactive**.

the sound isn't playing

Check these 5 things in order:

1. `index.html` must include `p5.sound`

Put this **after** p5.js:

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/
1.9.0/p5.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/
1.9.0/addons/p5.sound.min.js"></script>
```

If that second line is missing, sound will not work.

2. Your sound file must be in the right place

If you wrote:

```
spaceMusic = loadSound("space.mp3");
```

then `space.mp3` must be in the **same folder** as your `sketch.js`.

If it is inside another folder, use that path, for example:

```
spaceMusic = loadSound("assets/space.mp3");
```

3. Use `mousePressed()` to start audio

Browsers usually block autoplay, so the sound has to start after a click.

Use this:

```
function mousePressed() {
  userStartAudio();

  if (spaceMusic && !spaceMusic.isPlaying()) {
    spaceMusic.setVolume(0.5);
    spaceMusic.loop();
  }
}
```

4. Add a load callback so you know it actually loaded

Replace your preload line with this:

```
function preload() {
  worldData = loadJSON("world.json");
  spaceMusic = loadSound(
    "space.mp3",
    () => console.log("Sound loaded"),
    () => console.log("Sound failed to load")
  );
```

```
}
```

Then open the browser console.
If you see **"Sound failed to load"**, the file path is wrong or the file format is not working.

5. Use this full working `sketch.js` structure

Compare yours to this:

```
const VIEW_W = 800;
const VIEW_H = 480;

let worldData;
let level;
let player;
let camX = 0;
let camY = 0;

let spaceMusic;

function preload() {
  worldData = loadJSON("world.json");
  spaceMusic = loadSound(
    "space.mp3",
    () => console.log("Sound loaded"),
    () => console.log("Sound failed to load")
  );
}

function setup() {
  createCanvas(VIEW_W, VIEW_H);

  level = new WorldLevel(worldData);

  const start = worldData.playerStart ?? { x: 1200, y: 800,
speed: 1.8 };
  player = new Player(start.x, start.y, start.speed);

  camX = player.x - width / 2;
```

```
  camY = player.y - height / 2;
}

function draw() {
  player.updateInput();

  player.x = constrain(player.x, 0, level.w);
  player.y = constrain(player.y, 0, level.h);

  let targetX = player.x - width / 2;
  let targetY = player.y - height / 2;

  const maxCamX = max(0, level.w - width);
  const maxCamY = max(0, level.h - height);

  targetX = constrain(targetX, 0, maxCamX);
  targetY = constrain(targetY, 0, maxCamY);

  camX = lerp(camX, targetX, 0.04);
  camY = lerp(camY, targetY, 0.04);

  level.drawBackground();

  push();
  translate(-camX, -camY);
  level.drawWorld(player);
  player.draw();
  pop();

  level.drawHUD();
}

function mousePressed() {
  userStartAudio();

  if (spaceMusic && !spaceMusic.isPlaying()) {
    spaceMusic.setVolume(0.5);
    spaceMusic.loop();
    console.log("Tried to play music");
  }
```

```
}
```

Most likely reasons in your case

Usually it is one of these:

- `p5.sound.min.js` is not in `index.html`
- the file path to `space.mp3` is wrong
- you did not click the canvas before expecting sound
- the sound file format is unsupported/corrupted

Fastest test

Try a **different file**, preferably a small `.mp3`, and make sure it is in the same folder as `sketch.js`.

Paste your `index.html` and `preload()/mousePressed()` code here and I'll point to the exact problem.